

COS 226 – PRAC 5

GROUP MEMBERS

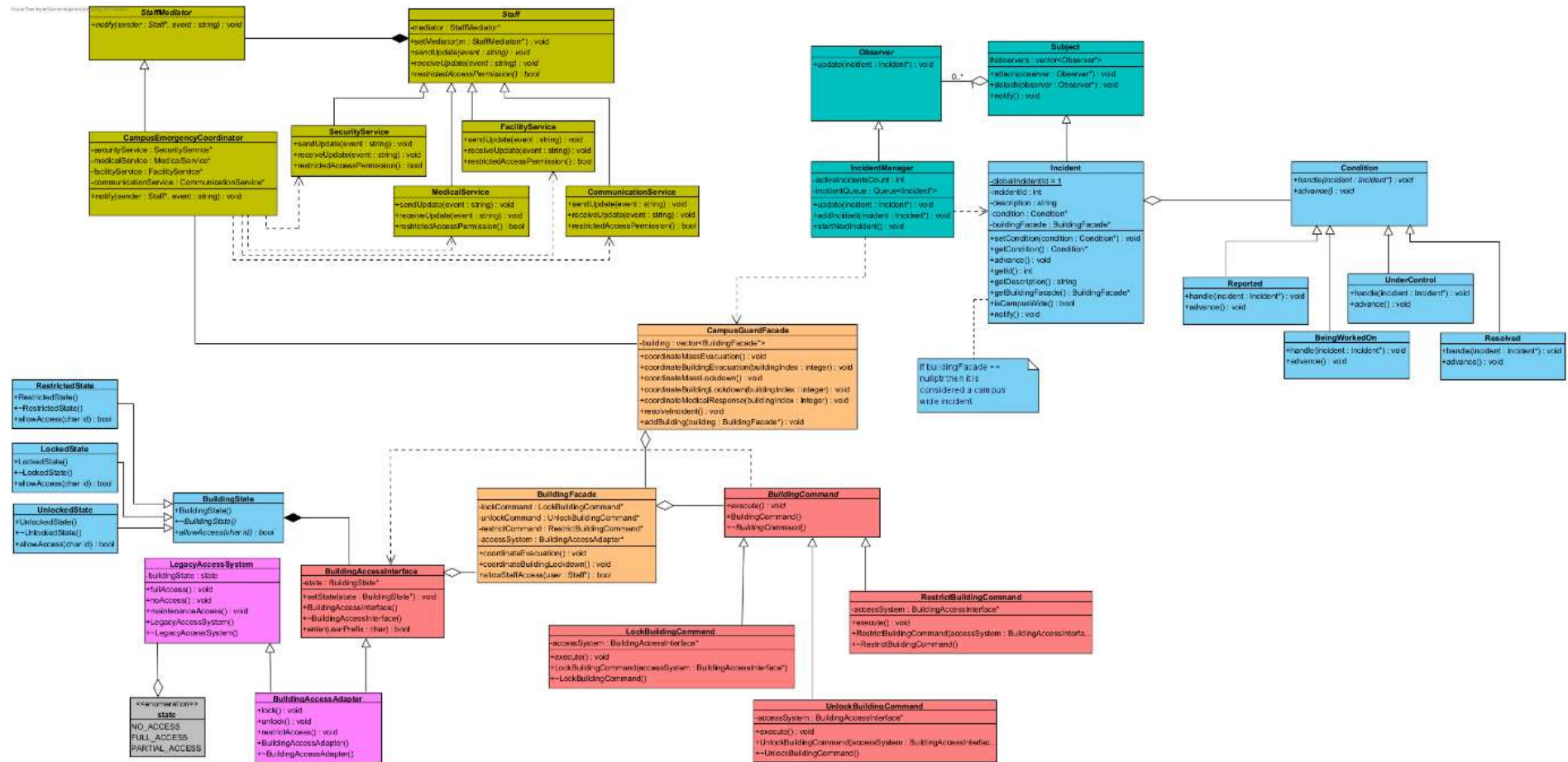
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CampusGuard description:

CampusGuard is an application designed to manage incidents that may occur on a campus by dispatching teams to resolve them and changing building access levels to prevent the public from entering restricted areas.

UML Class Diagram



Gof Participants

COMMAND

Invoker	BuildingFacade
Command	BuildingCommand
ConcreteCommand	LockBuildingCommand UnlockBuildingCommand RestrictBuildingCommand
Receiver	BuildingAccessInterface

MEDIATOR

Mediator	StaffMediator
ConcreteMediator	CampusEmergencyCoordinator
Colleague	Staff
ConcreteColleague	SecurityService
	MedicalService
	FacilityService
	CommunicationService

ADAPTER

Target	BuildingAccessInterface
Adaptee	LegacyAccessSystem
Adapter	BuildingAccessAdapter

STATE

Context	Incident
State	Condition
ConcreteState	Reported
	BeingWorkedOn
	UnderControl
	Resolved

OBSERVER

Subject	Subject
ConcreteSubject	Incident
Observer	Observer
ConcreteObserver	IncidentMangaer

FAÇADE

Facade	CampusGuardFacade BuildingFacade
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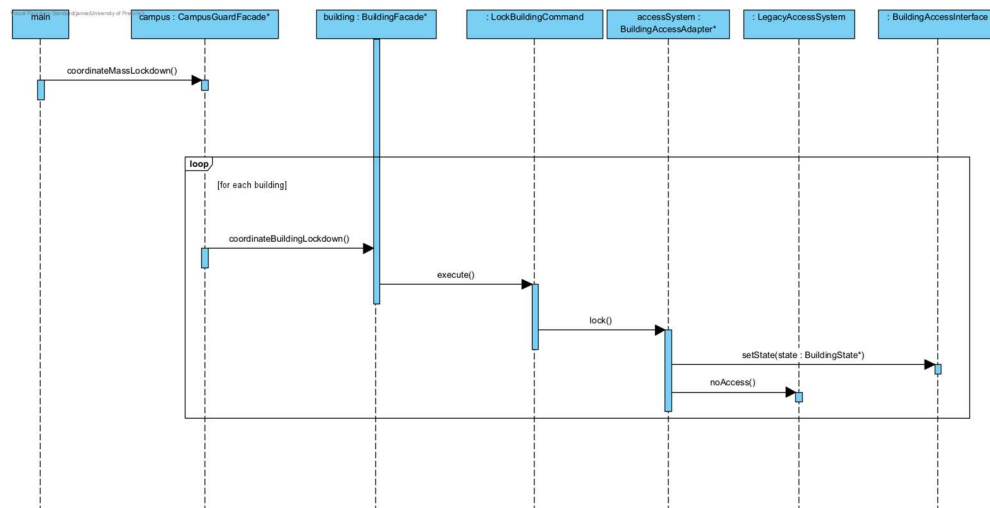
Team-selected patterns

State: We used the State pattern to keep track of the condition of the incidents which CampusGuard manages and resolves.

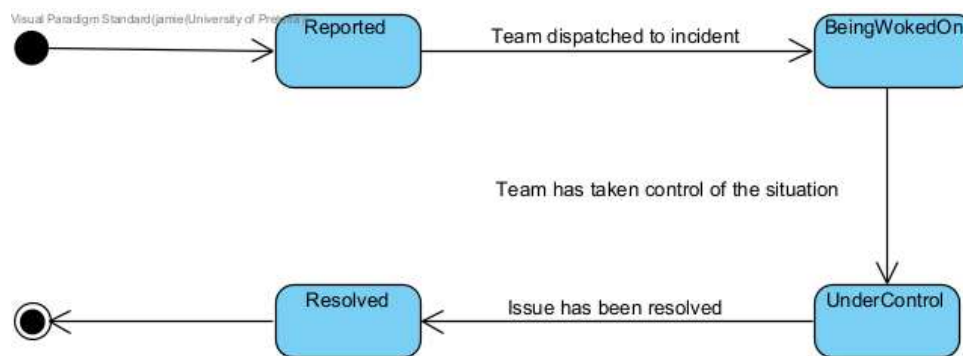
Observer: We used observer to keep track of the number of incidents currently being dealt with. There are a limited number of teams available to deal with the incidents, so when a team is finished with their incident it notifies the manager that they are now free so another may start being resolved.

Behavioural Diagrams

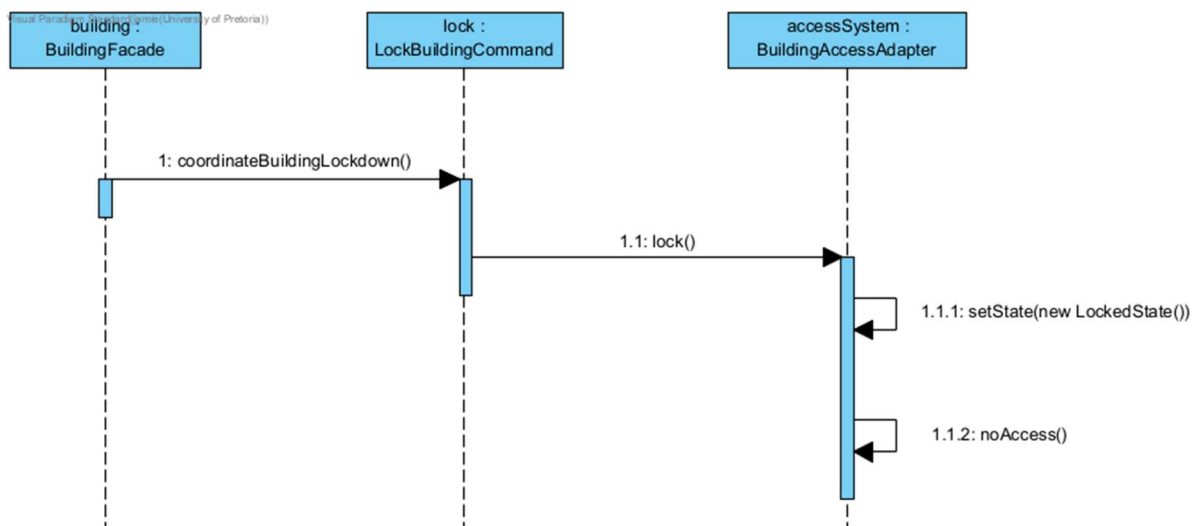
Mass Lockdown Sequence diagram



Incident Condition State diagram



Building Lockdown



Design Rationale

- We used a mediator for staff coordination, so that each colleague communicates through a centralized party in an emergency. If not used each colleague would have to contain a pointer to each other, making it complex and increasing coupling.
- Two-level Facade: BuildingFacade hides the Command/Adapter/State chain, while CampusGuardFacade keeps track of every building and can efficiently coordinate a mass evacuation or lockdown of every building on the campus.
- Ownership
 - BuildingAccessInterface has a compositional relationship with State and is responsible for deallocating its memory.
 - BuildingFacade has an aggregational relationship with CampusGuard. This is to allow the user to choose whether to use the façade or directly interact with the BuildingFacade. This also allows for scalability, meaning this system can work for an individual building only as well

Gdb Evidence

#1

Bug:

- Segmentation fault during mass evacuation.

Location:

- BuildingAccessInterface::setState()

Cause:

- Commands contained duplicate accessSystem pointers, hiding the initialized base-class pointer.

Fix:

- Removed duplicate accessSystem members from Lock/Unlock/Restrict commands.

Initial result:

- “104 bytes definitely lost in 13 blocks”.

```

--- Output/messages ---
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/usr/lib/x86_64-linux-gnu/libthread_db.so.1".
===== CAMPUSGUARD =====
There are 3 available teams ready to deal with incidents.
Fire detected in the IT Building.
Incident with id 1 has been added to the queue.
A staff member reports smoke coming from the IT Building.

Dispatching team to handle incident with ID: 1
Incident with id 1 has entered the following state: Being worked on.
There are 2 teams remaining on standby.

Active Incident:
Fire detected in the IT Building.

The incident is now being worked on.
Incident with id 1 has entered the following state: Under control.

CampusGuard initiates a mass evacuation.

=== MASS EVACUATION INITIATED ===

```

```

Program received signal SIGSEGV, Segmentation fault.
0x0000555555557fd8 in BuildingAccessInterface::setState(BuildingState*) ()
--- Assembly ---
0x0000555555557fd8 ? mov    (%rax),%rax
0x0000555555557fdb ? test   %rax,%rax
0x0000555555557fde ? je     0x555555557fef <_ZN23BuildingAccessInterface8setStateEP13BuildingState+47>
0x0000555555557fe0 ? mov    (%rax),%rdx
0x0000555555557fe3 ? add    $0x8,%rdx
0x0000555555557fe7 ? mov    (%rdx),%rdx
0x0000555555557fea ? mov    %rax,%rdi
0x0000555555557fed ? call   *__rdx
0x0000555555557fef ? mov    -0x8(%rbp),%rax
0x0000555555557ff3 ? mov    -0x10(%rbp),%rdx
--- Breakpoints ---
--- Expressions ---
--- History ---
--- Memory ---
--- Registers ---
rax 0x8786858483828180 rbx 0x000055555555778b0 rcx 0x000055555555778c0 rdx 0x000055555555638b0 rsi 0x000055555555778b0
rdi 0x8786858483828180 rbp 0x00007fffffd9bc0 rsp 0x00007fffffd9bb0 r8 0x0000000000000000 r9 0x00007ffff7a12ac8
r10 0x0000000000000020 r11 0x00007ffff7a12ac0 r12 0x0000000000000000 r13 0x8786858483828180 r14 0x00007ffff7fd000
r15 0x00005555555563740 rip 0x0000555555557fd8 eflags [ IF RF ] cs 0x00000033 ss 0x0000002b
ds 0x00000000 es 0x00000000 fs 0x00000000 gs 0x00000000 fs_base 0x00007ffff7f0780
gs_base 0x0000000000000000
--- Source ---
--- Stack ---
[0] from 0x0000555555557fd8 in BuildingAccessInterface::setState(BuildingState*)
[1] from 0x000055555555c9d5 in RestrictBuildingCommand::execute()
[2] from 0x0000555555558f54 in CampusGuardFacade::coordinateMassEvacuation()
[3] from 0x000055555555e780 in main
--- Threads ---
[1] id 1632 name campusGuard from 0x0000555555557fd8 in BuildingAccessInterface::setState(BuildingState*)
--- Variables ---
>>>

```

#2

After some modifications to the code and main, there was a memory leak in main

```

==5589==
==5589== HEAP SUMMARY:
==5589==   in use at exit: 16 bytes in 1 blocks
==5589== total heap usage: 66 allocs, 65 frees, 76,281 bytes allocated
==5589==
==5589== 16 bytes in 1 blocks are definitely lost in loss record 1 of 1
==5589==   at 0x4846FA3: operator new(unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==5589==   by 0x113A57: main (in /mnt/c/Users/sodhi/Documents/!School Stuff/Year 2/S2/COS 214/Pracs/COS214PA5/campusGuard)
==5589==
==5589== LEAK SUMMARY:
==5589==   definitely lost: 16 bytes in 1 blocks
==5589==   indirectly lost: 0 bytes in 0 blocks
==5589==   possibly lost: 0 bytes in 0 blocks
==5589==   still reachable: 0 bytes in 0 blocks
==5589==   suppressed: 0 bytes in 0 blocks
==5589==
==5589== For lists of detected and suppressed errors, rerun with: -s
==5589== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
sodhi@Stevie:/mnt/c/Users/sodhi/Documents/!School Stuff/Year 2/S2/COS 214/Pracs/COS214PA5$

```

To find which variable was undeleted, we first thought to check which pointers aren't null at the end. To do this, we set a breakpoint, then checked every pointer that was created

```
Breakpoint 1, main () at src/main.cpp:233
233      return 0;
(gdb) print fire
$1 = (Incident *) 0x0
(gdb) print zombie
$2 = (Incident *) 0x0
(gdb) print medicalEmergency
$3 = (Incident *) 0x0
(gdb) print library
$4 = (BuildingFacade *) 0x0
(gdb) print IT
$5 = (BuildingFacade *) 0x0
(gdb) print DiningHall
$6 = (BuildingFacade *) 0x0
(gdb) print person
No symbol "person" in current context.
(gdb) print person
No symbol "person" in current context.
(gdb) █
```

Person wasn't showing up. Then we realised that it must've went out of scope, since it was created in an if statement, so gdb couldn't see it, so we went and scanned the code, and saw we had not deallocated its memory, and did so.

```
155
156      CommunicationService *person = new CommunicationService();
157      std::cout << "\nCommunication Service Guy tries to enter." << std::endl;
158
159      if (building->allowStaffAccess(person))
160      {
161          std::cout << "Communication Service Guy allowed to enter." << std::endl;
162      }
163      else
164      {
165          std::cout << "Communication Service Guy not allowed to enter." << std::endl;
166      }
167
168      delete person;
169      person = nullptr;
170
171      active->advance();
```

And now there are no memory leaks

```
===== INCIDENTS SUCCESSFULLY RESOLVED =====
==6179==
==6179== HEAP SUMMARY:
==6179==    in use at exit: 0 bytes in 0 blocks
==6179== total heap usage: 66 allocs, 66 frees, 76,281 bytes allocated
==6179==
==6179== All heap blocks were freed -- no leaks are possible
==6179==
==6179== For lists of detected and suppressed errors, rerun with: -s
==6179== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

Valgrind Evidence

Cause:

State objects were being replaced without releasing the previous state.

Fixes:

- Added virtual destructor to Condition.
- Deleted previous state in Incident::setCondition().
- Deleted final state in Incident destructor.
- Added BuildingAccessInterface destructor to release the final BuildingState object.

Final result:

All heap blocks were freed -- no leaks are possible.

```
nare7@Nare:/mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5$ valgrind --leak-check=full ./campusGuard
==3289== Memcheck, a memory error detector
==3289== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==3289== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==3289== Command: ./campusGuard
==3289==
===== CAMPUSGUARD =====

There are 3 available teams ready to deal with incidents.
Fire detected in the IT Building.
Unauthorized access detected in Residence Hall.
Student collapsed outside the library.
Incident with id 1 has been added to the queue.
Incident with id 2 has been added to the queue.
Incident with id 3 has been added to the queue.

A staff member reports smoke coming from the IT Building.

Dispatching team to handle incident with ID: 1
Incident with id 1 has entered the following state: Being worked on.
There are 2 teams remaining on standby.

Active Incident:
Fire detected in the IT Building.

The incident is now being worked on.
Incident with id 1 has entered the following state: Under control.

CampusGuard initiates a mass evacuation.

=== MASS EVACUATION INITIATED ===
[Medical] Prepare emergency response
[Facility] Prepare evacuation
[Communication] Issue fire alert
```

```

===== INCIDENTS SUCCESSFULLY RESOLVED =====
==3289==
==3289== HEAP SUMMARY:
==3289==   in use at exit: 104 bytes in 13 blocks
==3289== total heap usage: 59 allocs, 46 frees, 76,017 bytes allocated
==3289==
==3289== 8 bytes in 1 blocks are definitely lost in loss record 1 of 13
==3289== at 0x485DFD3: operator new(unsigned long) (vg_replace_malloc.c:488)
==3289== by 0x400560F: Incident::Incident(std::_cxx11::basic_string<char, std::char_traits<char>, std::allocator<char> >) (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289== by 0x400A611: main (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289==
==3289== 8 bytes in 1 blocks are definitely lost in loss record 2 of 13
==3289== at 0x485DFD3: operator new(unsigned long) (vg_replace_malloc.c:488)
==3289== by 0x400560F: Incident::Incident(std::_cxx11::basic_string<char, std::char_traits<char>, std::allocator<char> >) (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289== by 0x400A696: main (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289==
==3289== 8 bytes in 1 blocks are definitely lost in loss record 3 of 13
==3289== at 0x485DFD3: operator new(unsigned long) (vg_replace_malloc.c:488)
==3289== by 0x400560F: Incident::Incident(std::_cxx11::basic_string<char, std::char_traits<char>, std::allocator<char> >) (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289== by 0x400A718: main (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289==
==3289== 8 bytes in 1 blocks are definitely lost in loss record 4 of 13
==3289== at 0x485DFD3: operator new(unsigned long) (vg_replace_malloc.c:488)
==3289== by 0x4008754: Reported::advance(Incident*) (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289== by 0x400573C: Incident::advance() (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289== by 0x4006AA7: IncidentManager::startNextIncident() (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289== by 0x400A7DF: main (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289==
==3289== 8 bytes in 1 blocks are definitely lost in loss record 5 of 13
==3289== at 0x485DFD3: operator new(unsigned long) (vg_replace_malloc.c:488)
==3289== by 0x400347A: BeingWorkedOn::advance(Incident*) (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)

```

```

==3289== at 0x485DFD3: operator new(unsigned long) (vg_replace_malloc.c:488)
==3289== by 0x400A1BC: UnderControl::advance(Incident*) (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289== by 0x400573C: Incident::advance() (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289== by 0x400A68: main (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289==
==3289== 8 bytes in 1 blocks are definitely lost in loss record 10 of 13
==3289== at 0x485DFD3: operator new(unsigned long) (vg_replace_malloc.c:488)
==3289== by 0x4008754: Reported::advance(Incident*) (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289== by 0x400573C: Incident::advance() (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289== by 0x4006AA7: IncidentManager::startNextIncident() (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289== by 0x400A803: main (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289==
==3289== 8 bytes in 1 blocks are definitely lost in loss record 11 of 13
==3289== at 0x485DFD3: operator new(unsigned long) (vg_replace_malloc.c:488)
==3289== by 0x400347A: BeingWorkedOn::advance(Incident*) (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289== by 0x400573C: Incident::advance() (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289== by 0x400A886: main (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289==
==3289== 8 bytes in 1 blocks are definitely lost in loss record 12 of 13
==3289== at 0x485DFD3: operator new(unsigned long) (vg_replace_malloc.c:488)
==3289== by 0x400A1BC: UnderControl::advance(Incident*) (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289== by 0x400573C: Incident::advance() (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289== by 0x400ABCF: main (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289==
==3289== 8 bytes in 1 blocks are definitely lost in loss record 13 of 13
==3289== at 0x485DFD3: operator new(unsigned long) (vg_replace_malloc.c:488)
==3289== by 0x400A270: UnlockBuildingCommand::execute() (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289== by 0x40052E7: CampusGuardFacade::resolveIncident() (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289== by 0x400AC09: main (in /mnt/c/Users/nare7/OneDrive/Documents/COS214/Prac/Prac5/COS214PA5/campusGuard)
==3289==
==3289== LEAK SUMMARY:
==3289==   definitely lost: 104 bytes in 13 blocks
==3289==   indirectly lost: 0 bytes in 0 blocks
==3289==   possibly lost: 0 bytes in 0 blocks
==3289==   still reachable: 0 bytes in 0 blocks

```

```

==3289==   still reachable: 0 bytes in 0 blocks
==3289==   suppressed: 0 bytes in 0 blocks
==3289==
==3289== For lists of detected and suppressed errors, rerun with: -s
==3289== ERROR SUMMARY: 13 errors from 13 contexts (suppressed: 0 from 0)

```

Collaboration Reflection:

Once we had the main design completed, we each would implement the different patterns by making a new branch for each pattern on github and merging once completed.

Contribution Statement:

Jamie:

- Made skeletons for all the classes
- Implemented Incident state pattern
- Implemented Observer pattern
- Set up dockerfile.
- Made State diagram

Stephen:

- Implemented building state pattern
- Implemented command pattern
- Wrote main file
- Made mass lockdown sequence diagram
- Made initial pdf mockup

Nare:

- Made initial class diagram
- Implemented mediator pattern
- Implemented Façade pattern
- Implemented adapter pattern
- Made building lockdown sequence diagram
- Performed gdb and valgrind testing